

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2024 Course: B. Tech. Sem. : V Branch: Mechanical Engg./Mechanical Engg. Sandwich Subject Code & Name: Heat Transfer (BTMC501) Max Marks: 60 Date: 01/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam table and Mollier chart is permitted.			
Q. 1	Attempt any two of the following	(Level/CO)	Marks
A)	Derive an equation for heat transfer rate through a hollow cylinder by applying Fourier's Law of Heat Conduction. The hollow cylinder has a length, L , inner radius as ' r_i ' and outer radius as ' r_o ', inner and outer wall temperatures as T_i and T_o . Thermal conductivity is represented as ' k '.	Understand	6
B)	An insulating powder is densely packed in the annular space between two concentric cylinders with radii 70 mm and 45 mm, of length 1m. The inner cylinder is uniformly heated with electric power input of 1 kW. Steady state temperature attained by inner sphere is 120°C and that by outer sphere is 30°C . Neglect thermal resistance offered by the material of the spheres. (i) Calculate thermal conductivity of the powder (ii) Calculate the thermal resistance offered by insulating powder.	Evaluate	6
C)	Explain the various modes of heat transfer with the help of neat sketches and expressions.	Remember	6
Q.2	Attempt any two questions A/B/C		
A)	A solid sphere 5 cm in diameter and initially at 500°C is quenched in a controlled environment at 90°C with convective heat transfer coefficient of $115 \text{ W/m}^2\text{K}$. Determine the time taken by centre of the sphere to reach a temperature of 150°C . Use the properties of the solid sphere as: C_p , Sp. heat = 418 J/kgK ; ρ , density = 7500 kg/m^3 , k , conductivity = 45 W/mK .	Analyze and Evaluate	6
B)	Write down the general differential heat conduction equation in Cartesian coordinate system for various situations for its use, e.g., Steady State, with no heat generation and one dimensional heat transfer.	Understand	6
C)	Calculate the fin efficiency and heat loss from a fin (assuming infinitely long fin), with circular cross section under following conditions: Air temperature : 30°C Temperature of fin at the base : 230°C Thermal conductivity of a fin : 200 W/mK Heat transfer coefficient over a surface of a fin : $15 \text{ W/m}^2 \text{ K}$ Fin Diameter : 10 mm	Understand	6
Q. 3	Attempt any two of the following		
A)	Sketch the laminar and turbulent boundary layers for a flow between two parallel plates.	Understand	6
B)	Discuss the fundamentals of various types of convection with the help of real life examples.	Understand and remember	6

C)	A hot rectangular plate 5 cm x 4 cm maintained at 200°C is exposed to still air at 30°C. Calculate the convective heat transfer rate by natural convection from both sides of the plate if smaller side of the plate is held vertical. Use following correlation: $Nu = 0.6 Ra^{(0.25)}$ Use the following air properties at Mean Film temperature of 115°C Density: 0.91 kg/m ³ , Cp = 1 kJ/kgK, Dynamic Viscosity (μ) 23×10^{-6} N-s/m ² , k = 0.033 W/mK.	Analyze and Evaluate	6
Q.4 Attempt any two of the following			
A)	Discuss the classification of Heat Exchanger with the help of neat sketches and practical examples.	Analyze	6
B)	In a double pipe counter flow heat exchanger 165 kg/min, of an oil having a specific heat 2100 J/kgK is cooled from 90°C to 60°C by 130 kg/min of cooling water at 25°C. Assume overall heat transfer coefficient of 400 W/m ² K and specific heat of water as 4200 J/kg-K. Find out LMTD and surface area of heat exchanger.	Analyze and Evaluate	6
C)	Discuss the terms: (i) Effectiveness of heat exchanger, (ii) NTU, (iii) Fouling factor of heat exchanger, (iv) Overall heat transfer coefficient	Understand	6
Q.5 Attempt any two of the following			
A)	Two large parallel plates, held at temperature 1000 K and 500 K, have emissivity values of 0.85 and 0.7, respectively. (i) Calculate the heat transfer rate by radiation per square meter of plate area (with no radiation shield), (ii) Calculate the heat transfer rate by radiation per square meter of plate area if the radiation shield having emissivity 0.5 is placed between the two plates (iii) Calculate the percent reduction in heat transfer rate by use of a radiation shield.	Analyze and Evaluate	6
B)	Derive an expression for radiation shape factor between two surfaces, separated by distance R. $F_{i-j} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{\cos\theta_i \cos\theta_j}{\pi R^2} dA_i dA_j$	Understand	6
C)	Define any three terminologies: 1) Stefan Boltzmann Law, 2) Kirchhoff's law, 3) Radiation shape factor, 4) Laminar and Turbulent flow	Understand	6
*** End ***			